

Database Structure

Overview

MongoDB stores the entire social graph, content layer, moderation workflows, rewards ledger, and livestream session state.

Core collections

Users

- model: `user.model.js`
- identity: username, email, phone, password
- location: country, verifiedCountry, detectedCountry
- roles: `role`, `roleName`, `accessRole`, `staffRole`, `publicRoles`
- social graph: followers/following
- economy: balances, wallet id, monthly reward counters
- auth state: refresh token, token revocation timestamps
- presence: `online`, `lastSeen`

Posts

- model: `post.model.js`
- author, community, text/media payload
- moderation state
- engagement counters
- `clientRequestId` for idempotent submission

Communities

- model: `community.model.js`
- country-scoped container for members and posts
- stores counts, moderators, and metadata

Notifications

- model: `notifications.model.js`
- sender, receiver, type, message, target routing data

Livestreams

- model: `LiveStream.js`
- host identity, Agora channel, lifecycle status, viewer counts, guest seats

Coin transactions

- model: `cointransaction.model.js`
- reward and transfer ledger
- ties wallet movement to posts, comments, live gifts, and system actions

Other notable collections

- `PostApproval`
- `ModerationItem`
- `Comment`
- `LikeActivity`
- `View`
- `Follow`
- `RoleActivationCode`
- `AppVerification`
- `MonthlyYkcSnapshot`
- monetization and ad metric collections

Known issues

- some view-specific fields are stored in core collections
- counters such as `memberCount` can drift from relationship reality
- role data is denormalized across several user fields

Scaling concerns

- several query paths rely on denormalized counters staying correct
- feed, moderation, and analytics routes use populate-heavy reads
- scheduled jobs scan broad collections without partitioning

Recommended improvements

1. document ownership of denormalized counters
2. add repair jobs or derived views for critical counters
3. define schema migration rules for legacy role fields